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KATMAP infers splicing factor activity and regulatory targets from knockdown data

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Typical RNA sequencing (RNA-seq) experiments uncover hundreds of splicing changes, reflecting underlying changes in splicing factor (SF) activity. Understanding how SF activity influences transcriptomic variation requires elucidating how each SF impacts splicing. Here, we present an interpretable regression model, KATMAP, which models splicing changes throughout the transcriptome by analyzing changes in SF binding and the resulting alterations in RNA processing. To learn a regulatory model, KATMAP requires SF perturbation RNA-seq data and the SF's binding motif as inputs, returning a description of the SF's position-specific regulatory activity and predicted targets. The KATMAP software includes models pretrained on ENCODE SF knockdown data. Learned KATMAP models can be applied to predict SF regulation and *cis*-elements at individual exons, which can guide the design of splice-switching antisense oligonucleotides. KATMAP can also interpret RNA-seq data by uncovering the factors responsible for transcriptomic changes, distinguishing direct SF targets from indirect effects and inferring relevant SFs from clinical RNA-seq data.

Programs of transcriptional and post-transcriptional regulation are responsible for development, physiology and disease. Orchestrating these gene expression programs are regulatory factors that recognize sequence elements distributed across the genome and transcriptome. The transcriptomic differences between two biological states are typically large, with hundreds or thousands of genes differing in splicing or expression, but these differences typically result from variation at a much smaller set of regulatory factors. Disentangling gene expression programs in terms of regulatory factors and their targets is critical to understanding the genomic basis of physiology and doing so requires determining the rules by which these factors exert their activities.

Alternative splicing greatly expands proteomic diversity by allowing a single gene to generate dozens of distinct splicing isoforms (spliceoforms), often with functions tuned to specific tissues or developmental stages. Splicing can also impact gene expression levels as some spliceoforms may be subject to turnover through pathways such as nonsense-mediated mRNA decay. Approximately 10–20% of disease-causing mutations are estimated to result in changes to splicing levels of individual exons[1,2], while perturbations of splicing factor (SF) genes can broadly disrupt splicing. Several SFs are known oncogenes or tumor suppressors[3,4] and the resulting disruptions to splicing programs represent vulnerabilities that can be exploited[5]. In a therapeutic context, splicing defects at individual exons can often be corrected by splice-switching antisense oligonucleotides (ssASOs)[6,7]. However, brute-force tiling is often required to identify effective ASOs because the relevant SFs and splicing regulatory elements (SREs) are unknown. Interpretation of clinical variants that may impact splicing is also made challenging by our incomplete knowledge of SF activities and SREs[8]. Elucidating precise rules governing SF activity would facilitate variant interpretation, ASO design and the identification of SFs that contribute to disease.

A number of exploratory approaches have been developed to assess how the presence of particular motifs or *k*-mers are associated with variation in transcription[9,10], mRNA stability[11] or splicing[12,13]. Because these models require discovering the putative activity of hundreds or thousands of motifs and *k*-mers every time they are applied, they do not incorporate prior knowledge of each factor's regulatory activity, extract generalizable insights or even directly identify the causal factors. More focused approaches have leveraged perturbation experiments to predict the targets of specific SFs by incorporating

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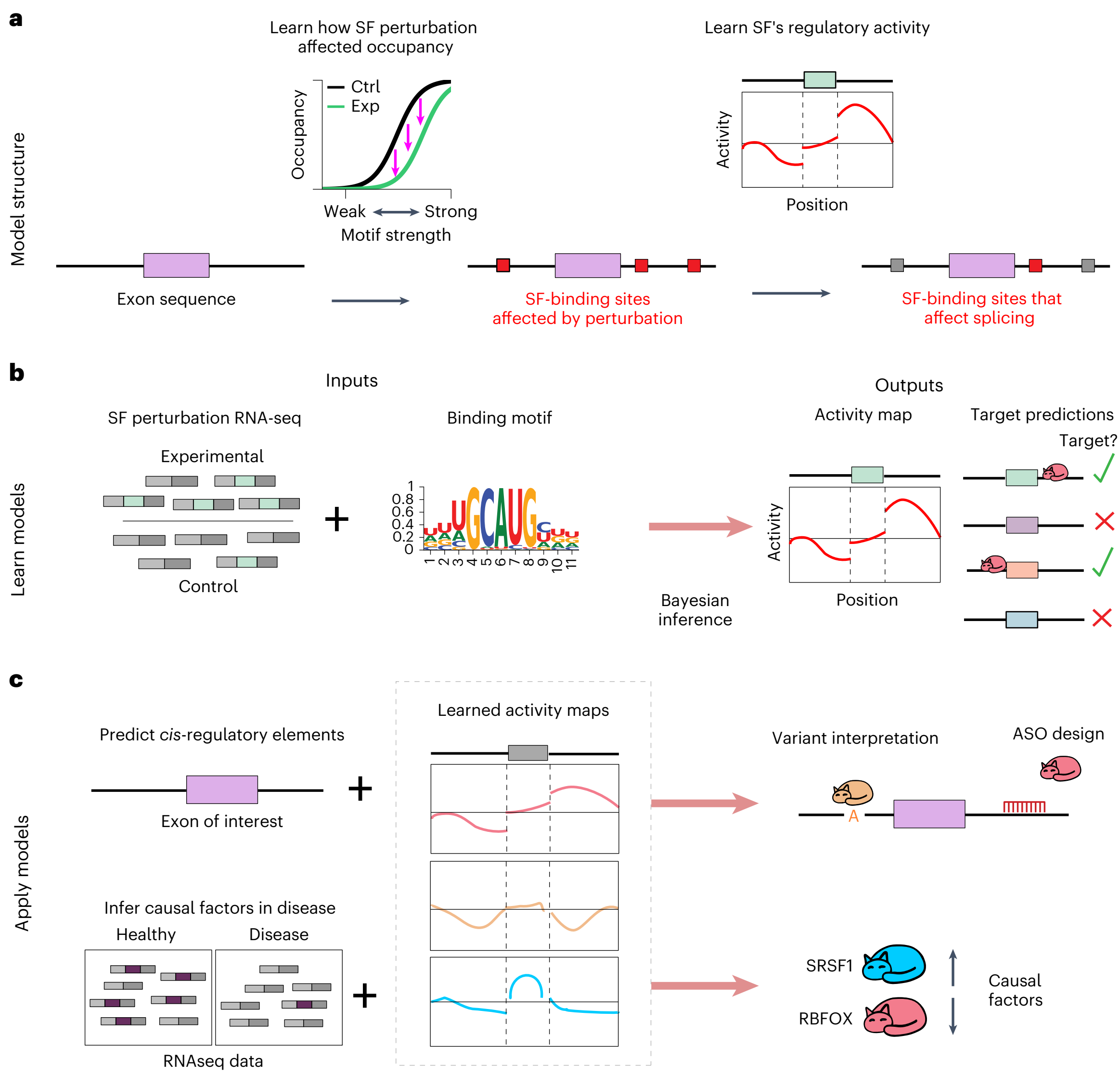

Fig. 1 | High-level overview of the KATMAP model and its applications. a, The general workflow of how KATMAP predicts the regulation of individual exons on the basis of their sequence. **b**, Learning the rules by which an SF regulates splicing requires only an RNA-seq dataset where that SF was perturbed and some model of its sequence specificity. Bayesian inference determines which splicing models are consistent with the observed splicing changes, yielding an activity map describing the SF's position-specific regulatory activity and identifying which exons are the SF's direct targets and which splicing changes represent indirect effects. **c**, Examples of how learned models can be applied to interpret splicing in terms of SF regulation. Top, scoring an exon of interest to predict the SREs and SFs that control its splicing, with application to variant interpretation and ASO design. Bottom, identifying the causal SFs underlying splicing differences observed in a given RNA-seq dataset.

expert knowledge of the factor and using crosslinking and immunoprecipitation (CLIP) data to inform binding sites[14–16]. However, the challenge of crafting these models to the specific details of the SF's binding and activity has limited their application to a few well-studied factors and a reliance on crosslinking data still ties the insights to exons where the SF was detectably bound in specific experimental data. This need to tailor models to the specifics of particular SFs has been sidestepped by recent applications of deep learning to the prediction of splicing from the nucleotide sequence[2]. However, the learned models are black boxes containing myriad parameters representing abstract transformations rather than biological statements. Follow-up in silico experiments are typically required to predict *cis*-regulatory elements or the associated positional activities[17].

Here, we present a flexible and interpretable model for learning regulatory activity from knockdown or overexpression RNA-seq data, which can then be applied to generate predictions and interpret sequencing data. Our approach, knockdown activity and target models from additive regression predictions (KATMAP), explicitly models the effect of knockdown by assigning changes in SF occupancy at potential binding sites throughout the transcriptome to predict regulation at individual exons (Fig. 1a). We use a hierarchical framework that tailors modeling assumptions to each SF and learns the nonlinear biophysical relationship between motif affinity and binding, allowing KATMAP to be broadly applied off-the-shelf to diverse SF datasets. Learning a regulatory model requires as inputs only the results of a perturbation experiment and some knowledge of the factor's binding motif, such as a position-weight matrix (PWM) (Fig. 1b). The learned model describes the position-dependent regulatory activity for the SF of interest. Once learned, these models can be applied to any exon of interest to identify *cis*-elements and associated SFs regulating its inclusion, which can simplify ASO design, or to any RNA sequencing (RNA-seq) dataset to infer which SFs explain the splicing differences observed between conditions (for example, to identify disease drivers; Fig. 1c).

Results

Overview of the KATMAP model

KATMAP provides a framework for learning interpretable models of splicing regulation from perturbation experiments, which can be applied to interpret splicing regulation in other datasets. To learn a regulatory model from a perturbation experiment, KATMAP uses a model of the SF's binding motif to explain the splicing changes uncovered in a differential splicing analysis of the perturbation experiment (Fig. 1b). The differential splicing calls can be defined with the user's software of choice, although we use rMATS results[18] in our analyses (instructions for organizing the results of MAJIQ v2 (ref. 19) are also provided in the documentation). Sequence specificity can be represented with a PWM or position-specific affinity matrix (PSAM), a collection of weighted PWMs and PSAMs or a ProBound model[20]. The PWMs may be derived from in vitro binding data, eCLIP (for example, with MCROSS[21]) and, for full interpretability, should represent the binding specificity of the SF of interest or a close homolog. Motifs discovered de novo from the RNA-seq data (for example, with MEME-suite) can also be used as input. Of course, motif discovery analyses may identify motifs reflecting secondarily perturbed SFs; hence, the models learned using de novo motif discovery will not be as interpretable as when the motif is known.

KATMAP infers which motifs were affected by the perturbation experiment and distills the observed splicing changes into a description of the SF's position-specific regulatory activity, which we term an activity map (Fig. 1a,b). This learned model provides a scoring system for predicting whether an exon is regulated by an SF on the basis of (1) whether there are binding motifs of appropriate strength and (2) whether these motifs are located in exon-proximal regions where the SF exerts regulation. These criteria allow KATMAP to predict which splicing changes reflect the direct targets of the perturbed SF and which are the indirect effects of perturbation (Fig. 1b). Focusing on an exon of interest, our models can be used to predict which SFs regulate that exon and identify the corresponding *cis*-elements that are bound (Fig. 1c). Alternatively, these models can be applied to infer which SF perturbations underlie the splicing profile of an RNA-seq dataset relative to a control by asking whether the exons with altered splicing overlap with the targets of particular SFs. The KATMAP python library (with pretrained models) is available from GitLab (https://gitlab.com/LaptopBiologist/katmap) and provides a unified framework for these applications.

Architecture of the regression model

The intuition guiding our model is as follows. Before a perturbation to the activity of a sequence-specific SF, sequence motifs in the transcriptome are bound by the SF at varying levels of occupancy. We assume the factor can only influence the splicing reaction if its binding sites are properly situated relative to the exon and that binding elsewhere has negligible effect. After knockdown, the free protein concentration decreases, leading to reduced occupancy across the protein's binding sites. If binding is lost from a regulatory region, then loss of regulation results in a change in exon inclusion. The greater the decrease in binding, the more likely it is to affect inclusion. The logic is similar for SF overexpression, except that free protein concentration, occupancy and regulation increase upon perturbation.

Following this logic, KATMAP models splicing changes in terms of changes in SF binding to the transcript. As observations, we use the results of a differential splicing analysis that has labeled exons as upregulated, downregulated or not significantly changed. We score the sequence around each of these exons using a model of the factor's sequence specificity. KATMAP aims to predict whether and in which direction each exon changes in splicing by answering two questions: (1) Are there binding sites near the exon that were affected by the SF knockdown? (2) Where must the SF bind near an exon to regulate its inclusion? We frame this as a generalized additive model that represents SF binding and regulation with nonlinear terms but accounts for other features predictive of detectable splicing change (such as the splicing event's read count) through additional linear terms.

Our model begins by scoring all potential binding motifs around an exon's splice sites (SSs), extending 200 nt on the intronic and 60 nt on the exonic sides, using a model of the SF's motif (Extended Data Fig. 1a). These *motif scores* are then transformed to changes in binding using a binding function based on a biophysical description of how occupancy changes after a decrease in SF availability (Extended Data Fig.1b,c). The shape of the binding function is controlled by a 'most relevant affinity' parameter that defines which binding sites were most affected by the knockdown (that is, shifts the function along the *x* axis) and an 'affinity scale' parameter that defines the range of motif scores affected by the knockdown (that is, widens or narrows the binding function); we refer to these collectively as the binding parameters (Methods).

We represent the SF's position-specific regulatory activity with a set of parameters termed the activity map (Extended Data Fig. 1d), which assigns enhancing (>0) or suppressing (<0) regulatory activity to different positions on the intronic and exonic sides of 3′ and 5′ SSs (3SS and 5SS). These regulatory activities are shared across all exons in the analysis. Critically, we assume that activity varies smoothly with position, with discontinuities at the exon–intron boundaries that allow exonic and intronic binding to have distinct activities. The degree of smoothness is learned from the data, rather than assumed a priori, in the form of two spatial correlation parameters. We multiply each change in binding by the corresponding activity to determine its impact on the splicing outcome (Extended Data Fig. 1e). The sum of these products defines the SF's splicing impact on the exon, making the simplifying assumption that regulation is additive.

Lastly, we incorporate these splicing impacts into a regression framework that assigns each exon a probability of being upregulated, downregulated or unchanged (Extended Data Fig. 1f). In addition to SF binding, the unperturbed level of exon inclusion and gene expression impact our statistical power to detect splicing changes. Specifically, exons with inclusion levels (percent splicing inclusion (PSI) values) near the boundaries of 0% or 100% inclusion are less likely to be detected as downregulated or upregulated, respectively, while statistical power to detect splicing changes is limited for exons in genes with low expression. We include these confounding variables (for example, unperturbed inclusion level and expression) and their squares as linear predictors, whose effects are controlled for by a set of linear coefficients.

This model architecture answers the two questions we posed in an integrated fashion. SF binding and regulatory activity of the perturbed SF are learned simultaneously, without the need to define arbitrary cutoffs for binding or activity. The activity map provides a simple description of the SF's position-specific activity, identifying where a binding site must occur to influence an exon's inclusion positively or negatively, which we propose as an alternative to the traditional RNA map.

The goal of KATMAP's inference algorithm is to determine which parameter values provide good explanations for the observed splicing changes. Prior target prediction models obtained point estimates of the most likely parameter values[14], but we wished to also quantify our uncertainty by identifying the range of plausible hypotheses. We, therefore, used Bayesian inference to precisely evaluate the plausibility of hypotheses given observed data (Extended Data Fig. 1i), which we accomplish by coupling adaptive importance sampling[22] with integrated nested Laplace approximations (INLA)[23]. This allows us to compute point estimates of the most plausible parameters, quantify our uncertainty with credible intervals (CIs) and propagate this uncertainty forward into our predictions of regulation. Finally, the learned parameters can be used to score new sequences.

KATMAP accurately infers parameters and quantifies uncertainty

Before applying our model to interpret experimental data, we wanted to confirm that our inference strategy could correctly infer regulatory activity. We simulated 21 SF knockdown datasets with known parameter values and found that KATMAP robustly recovers the true parameters (Extended Data Fig. 2a–c). Importantly, our posterior samples properly quantify the uncertainty regarding the activity and binding parameters (Extended Data Fig. 2e,f). Because the activity and binding parameters are accurately recovered, so are the splicing impacts used to simulate the splicing changes (Extended Data Fig. 2d). The linear coefficients are slightly underestimated (Extended Data Fig. 2c), meaning that their true values are less frequently captured by the CIs than expected (Extended Data Fig. 2g), likely because of the multivariate Gaussian approximation to their conditional posterior[23]. Nonetheless, the accurate recovery of the activity and splicing impacts shown above indicates that this subtle bias does not meaningfully interfere with the primary function of the linear coefficients in controlling for potential confounders.

To assess performance on real data and ensure that the models had not overfit, we selected 15 knockdowns with many splicing changes. For each experiment, we trained on 70% of the differentially spliced exons, holding out the remaining 30% as part of the test set. We evaluated our models against these held-out exons, comparing the true splicing changes to our predictions of which exons were upregulated, downregulated or unchanged upon knockdown. We found that the predicted probabilities were well calibrated (Extended Data Fig. 2h) and effective at predicting the splicing changes in the held-out exons (Extended Data Fig. 2i).

KATMAP robustly infers splicing regulatory activity

KATMAP aims to distill perturbation experiments into practical descriptions of the perturbed SF's splicing regulatory activity. Here, we applied KATMAP to knockdown data from the ENCODE project for 35 SFs for which we also had good models of the protein's sequence specificity (Supplementary Table 1)[24]. We used RNA-bind-n-seq-derived PSAMs as the binding model for 23 SFs[25,26] and RNAcompete-derived PWMs for the remaining 12 (refs. 27,28). A total of 18 SFs showed significant enhancing or repressing activity maps and outperformed a reduced model (with splicing activity excluded), with nine displaying enhancing activity, six showing repressing activity and three with both enhancing and repressing activity (Fig. 2a). Among these 18, 14 had available knockdown/RNA-seq data from both lymphocyte-derived K562 cells and hepatocyte-derived HepG2 cells. For all but one factor (SF1), the activity maps were consistent between the two knockdowns, highlighting the reproducibility of our inferences (Fig. 2b).

The evidence for splicing activity correlated with knockdown efficiency ($r = 0.45$; Extended Data Fig. 2j) and enhancing or repressing activity was only confidently detected in knockdowns with at least 100 downregulated or upregulated exons, respectively (Extended Data Fig. 2k–m). For example, both ENCODE knockdowns for the splicing activator TRA2A yielded <100 downregulated exons and no clear evidence of activity, perhaps because of compensation by paralog *TRA2B*. However, applying our model to a *TRA2A/B* double-knockdown experiment with >1,000 downregulated exons[29] recovered strong evidence of the expected exonic enhancing activity (Fig. 2c).

Examining the significant activity maps identifies three distinct groups of splicing activators. First, we identify SRSF1, SRSF7 and TRA2A/B as exhibiting the classic exonic enhancing pattern of SR protein regulation[30] (Fig. 2a,c). TRA2A/B shows similar activity near both SSs, while SRSF1 is inferred to have stronger activity near the 5SS and SRSF7 near the 3SS. Second, several polypyrimidine-binding factors show enhancing activity directly upstream of the 3SS, consistent with polypyrimidine tract (PPT) definition. For U2AF2 (which canonically defines the PPT[31]) and its homolog PUF60 (ref. 32), we infer intronic enhancing activity just upstream of the 3SS (Fig. 2a,d). U2AF2's inferred activity is constrained to within 20 nt of the 3SS, while PUF60's activity is slightly more upstream and extends 40–60 nt into the intron, consistent with observations of extended PPTs at PUF60-responsive exons[33]. Similar upstream enhancing activity is inferred for PCBP1 (Fig. 2a) but with additional activity downstream of the 5SS, consistent with prior observations of C-rich motifs upstream and downstream of exons activated by PCBP1 and PCBP2 (ref. 34). KATMAP failed to detect the expected enhancing activity of the branch point-binding factor SF1 (ref. 35) and the exonic enhancing activity of SRSF5, perhaps because of indirect effects of other SFs obscuring the signal in these knockdowns.

While upstream activators display greatest activity near the 3SS, a third group of factors with downstream enhancing activity (DAZAP1, HNRNPF/H, QKI and RBFOX) have their greatest activity tens of nucleotides away from the 5SS and influence splicing over a wider intronic window (Fig. 2a). Among these, two pairs of factors have strikingly similar activities. The related poly(G)-binding factors HNRNPF and HNRNPH1 both enhance inclusion when bound in the downstream intron while suppressing inclusion in the exon. HNRNPH1 was one of the first SFs demonstrated to activate and repress inclusion in a position-dependent manner[36,37]. RBFOX2 and QKI also share patterns of activity (Fig. 2e,f), enhancing downstream while suppressing upstream; however, unlike HNRNPF/H, they are not related and bind distinct motifs (YGCAUG versus ACUAAC, respectively).

In contrast to splicing activators, all inferred repressors displayed activity in multiple regions (Fig. 2a). The PCBP-related HNRNPK protein shows silencing activity upstream of the 3SS and in the exon (Fig. 2g), consistent with its role as a splicing repressor[38] and with observations that PCBP2 and HNRNPK can regulate the same exon in opposing directions[39]. We find a similar pattern of upstream and exonic silencing activity for PTBP1, as well as evidence of downstream enhancing activity (Fig. 2a), all of which is consistent with prior studies[40]. This exonic activity of HNRNPK and PTBP1 stands in contrast to the pyrimidine-binding upstream activators, which lacked exonic activity. The other repressors have even wider regions of suppression. HNRNPC's silencing activity is strongest in the ~100 nt upstream and downstream of the exon but has two additional regions of silencing activity at ±200 nt (Fig. 2h). HNRNPC is known to form tetramers, with repression suggested to result from looping out the exon, which may explain the multiple regions of activity[41]. HNRNPL activity maps also display repressing activity both upstream and downstream (Fig. 2a) but also indicate repression within the exon, consistent with prior work[42,43]. MATR3 (Matrin3) has even broader regions of repression, extending throughout both upstream and downstream introns in the HepG2 model but only showing evidence of upstream repression in K562 cells (Fig. 2a). This broad silencing activity is consistent with CLIP-based enrichment maps of MATR3-repressed exons, which found MATR3 uniformly distributed hundreds of bases into the flanking introns[44].

While we focus on cassette exons, KATMAP's statistical framework is general enough to accommodate other classes of splicing events. To explore this, we modeled regulation of alternative 3SS and 5SS choice, predicting usage of the SS distal to the constant exonic region on the basis of SF-binding sites in the constant intronic region and the variable exon segment between the two alternative SSs (Extended Data Fig. 3a). The evidence for splicing regulation inferred from alternative SSs was typically weaker than observed for cassette exons, perhaps because the inferences were often based on fewer significant events and shorter sequence regions. While fewer models satisfied our model comparison criterion, the activity maps showing significant enhancing or repressing activity were typically consistent with those learned from cassette exons (Extended Data Fig. 3b–n). This demonstrates that KATMAP can model alternative SS choice and suggests that our model comparison criterion may be conservative. The interpretation of activity in the variable region differs from the cassette exon models, with positive values favoring inclusion of the variable region as an exon (either by enhancing the distal SS or repressing the proximal SS) and negative values favoring usage of the variable

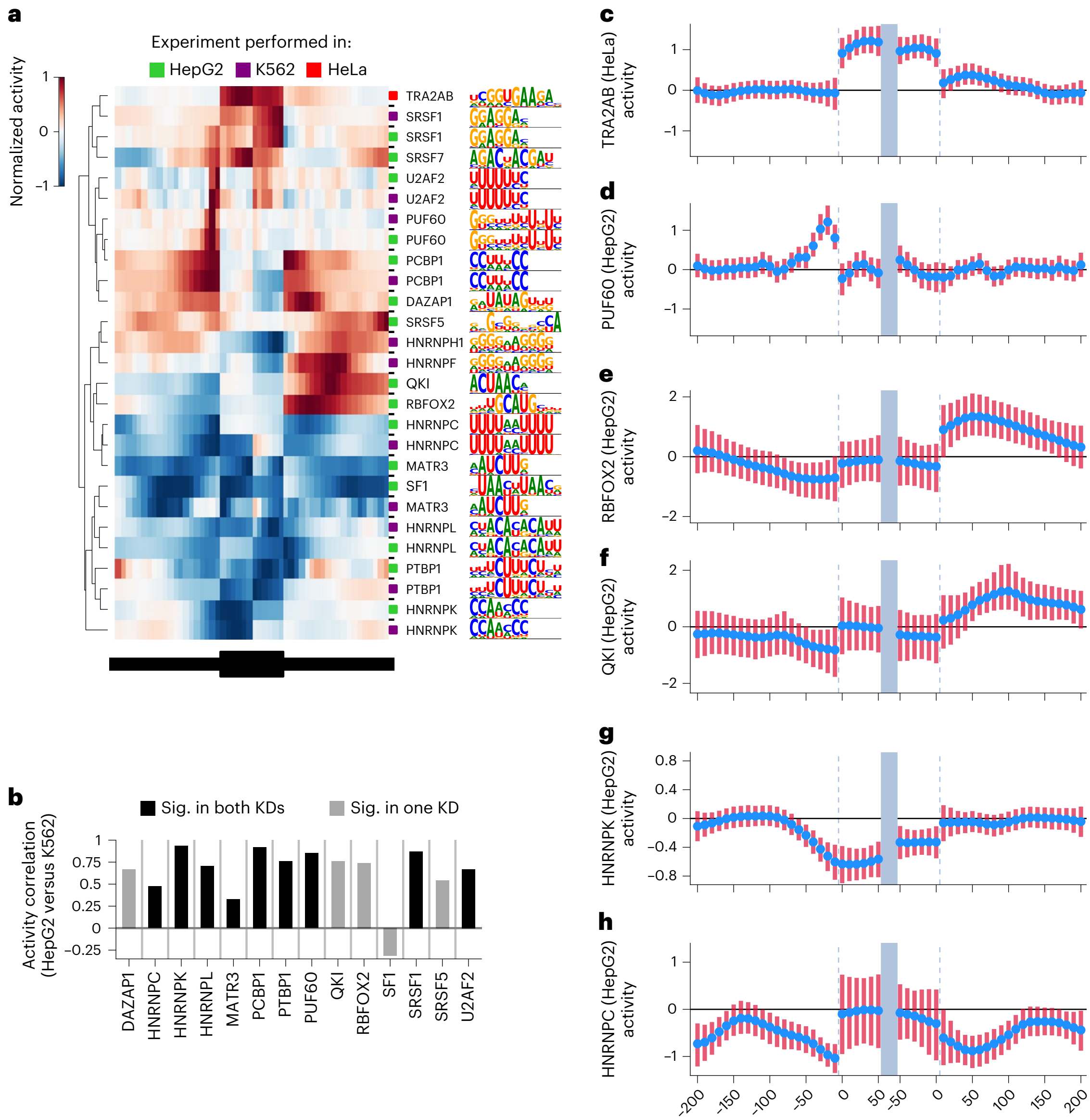

Fig. 2 | KATMAP robustly infers splicing activity. a, Significant activity maps and their motifs organized by hierarchical clustering. For ease of comparison, activity is normalized relative to the maximum absolute activity in each map. If the binding model included multiple motifs, only the top motif is shown. **b**, The correlation between activity maps for the same RBP inferred from different knockdowns. KD, knockdown. **c**–**h**, Activity maps for TRA2A/B (**c**), PUF60 (**d**), RBFOX2 (**e**), QKI (**f**), HNRNPK (**g**) and HNRNPC (**h**). The blue dots indicate the posterior mean and the lines are 94% CIs (n = 1,682, 1,420, 1,024, 1,106, 1,047 and 1,122 posterior samples, respectively).

region as an intron. For example, binding of PCBP2 and SF1 in the variable region between alternative 3SSs favors its usage as an intron (Extended Data Fig. 3i,j), likely reflecting activation of the proximal 3SS. This is consistent with the intronic enhancing activity observed for PCBP1 in the cassette exons maps (Fig. 2a) and SF1's known role in branch point definition[35].

The diverse position-dependent patterns of splicing regulation we uncover underscore the power and flexibility of KATMAP's approach to describing activity. Our assumption that this activity changes smoothly with position (except at exon–intron boundaries) is biologically plausible[45] and learning this smoothness from the data (rather than assuming this a priori) improved statistical power by pooling information spatially (Extended Data Fig. 1g,h) and allowed KATMAP to detect both broad and narrow ranges of activity. We were able to robustly and reproducibly uncover this activity by modeling the sequences of exons and introns without using crosslinking data, avoiding the bias toward exonic signals of eCLIP-based RNA maps[24].

Activity maps define regulatory targets

The changes observed in a differential splicing analysis after knockdown are a mixture of direct SF targets and the indirect effects of the perturbation. When studying an SF of interest, these indirect effects can obscure the underlying biological signal. However, the activity maps inferred by KATMAP lead to a natural definition of an SF's direct targets as exons with motifs of appropriate strength and location for regulation, distinguishing these from exons that lack binding motifs in regions expected by the model to impact inclusion (predicted nontargets). This is summarized in the 'splicing impact' score assigned to each exon, computed by multiplying predicted changes in binding by the SF's activity map (Extended Data Fig. 1c–e and Supplementary Table 2). We consider an exon to be a direct repression or enhancement target if the splicing impact of the exon is confidently nonzero and large enough to improve predictions over a reduced model that excludes all splicing activity (Fig. 3a). Exons not predicted to be targets reflect several possibilities. Some predicted nontargets should reflect exons

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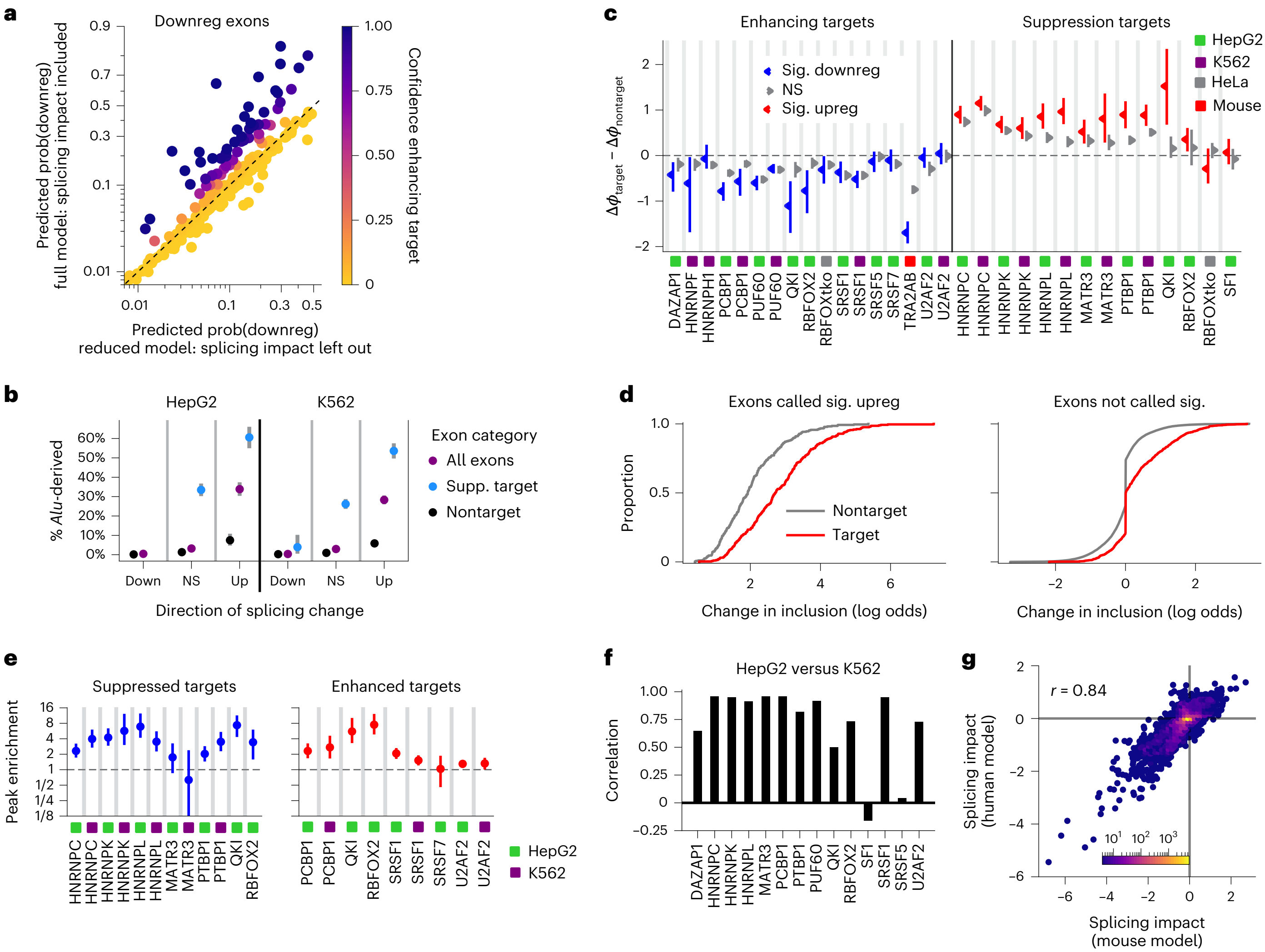

Fig. 3 | KATMAP predicts regulatory targets. a, The probabilities of downregulation predicted by the full model and a reduced model lacking splicing impacts for exons downregulated upon RBFOX2 knockdown (HepG2). Exons where the full model assigns a higher probability than the reduced model are likely enhancing targets, while those assigned the same predictions by both models are likely nontargets. The axes are in log odds scale and depicted in the range 0.01–0.99. **b**, The fraction of exons derived from *Alu* among targets of HNRNPC's suppression, nontargets or all exons that changed in a given direction, computed for each of the three splicing categories, with the posterior means and 94% CIs. **c**, The difference in average inclusion level change between confident targets and confident nontargets among exons that were significantly altered by knockdown (red and blue) and those that were not (gray). NS, not significant. The 94% CIs were obtained by Bayesian bootstrapping ($n = 2{,}000$). **d**, eCDF plots of the distributions of inclusion level changes at targets and nontargets upon HNRNPC knockdown (HepG2) for exons significantly upregulated upon knockdown (left) and nonsignificant exons (right). It is the difference in the means of the target and nontarget distributions that is shown in **c**, where we excluded exons where the data supported a confidently small effect of knockdown (spike at 0, $|\Delta\psi| < 10^{-3}$) to focus on those exons called nonsignificant potentially because of a lack of statistical power; the estimates with all exons included are shown in Extended Data Fig. 4b. **e**, The enrichment of eCLIP peaks in differentially spliced targets compared to nontargets differentially spliced in the same direction, with 94% CIs computed under a beta-binomial model ($n = 2{,}000$ posterior samples). The enrichment summarizes eCLIP peaks that span the significant regions of a given SF's activity map. **f**) The correlation in splicing impacts of exons across the human transcriptome computed from the HepG2 and K562 models for SFs with data for both cell types. **g**, Comparing RBFOX splicing impacts in the mouse transcriptome computed using the human RBFOX2 (HepG2) model and a model inferred from mouse *Rbfox1/2/3* triple-knockout data.

truly not bound or regulated by the SF (that is, true nontargets), while others might reflect true direct targets where the nature of binding or regulation is more complex than that assumed by the model or where the knockdown/RNA-seq data did not provide sufficient power for confident identification. To investigate these possibilities, we compared our target predictions to the effect sizes of knockdown on inclusion, crosslinking data and prior literature.

Comparing these target predictions to the literature recovers previously described targets. For example, PTBP1 is known to silence exons in its paralogs *PTBP2* (ref. 46) and *PTBP3* (ref. 47) during development, as well as in *RTN4* (ref. 48) and *FLNA*[49], all of which were identified as direct targets of PTBP1 by KATMAP. To assess our target predictions more comprehensively, we considered HNRNPC, which is known to silence exons derived from the highly abundant transposable element *Alu*[50]. These *Alu*-derived exons provide a ground truth for validating our target predictions. Intersecting predicted HRNPNC targets with the UCSC RepeatMasker track (https://www.repeatmasker.org/) confirmed that roughly 30% of all exons upregulated upon HNRNPC knockdown are *Alu*-derived. When we divide these into predicted targets and nontargets, ~55% of predicted targets are derived from *Alu* compared to only ~5% of nontargets. Thus, KATMAP's target predictions recover HNRNPC's known silencing of *Alu* exons, strongly sorting *Alu*-derived

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exons into the set of targets, with few labeled as nontargets (Fig. 3b), supporting the model's classifications of direct versus indirect targets.

We then assessed the extent to which our target and nontarget predictions correspond to the sensitivity of exons to the SF's knockdown. While our models were trained to predict the directions but not the magnitudes of changes in exon inclusion, among differentially spliced exons, those predicted to be direct targets had larger changes in inclusion following knockdown than predicted nontargets (Fig. 3c). For example, among exons upregulated upon HNRNPC knockdown, those with larger splicing impacts tended to have larger changes in inclusion (Fig. 3d and Extended Data Fig. 4a), with the average effect of knockdown being twofold higher at upregulated predicted HNRNPC targets compared to upregulated predicted nontargets. KATMAP's predicted targets also include exons that were not called significant in the differential splicing analysis but have binding motifs properly situated to affect splicing. These may represent true targets where the RNA-seq coverage was inadequate to confidently detect a splicing change. Consistent with this, 'nonsignificant' exons predicted to be targets on average changed more in the expected direction than predicted nontargets (Fig. 3c,d and Extended Data Fig. 4a). KATMAP's target predictions can, therefore, help interpret marginal signals in differential splicing analyses, using the presence of binding motifs to flag the subset most likely to be regulated by the SF.

KATMAP target predictions are based on sequence similarity to motifs identified in vitro and do not consider more complex binding modes, such as RNA structural preferences. To assess whether predicted targets were bound by the analyzed SF, we used crosslinking data. For the 11 SFs with credible activity maps that also had ENCODE eCLIP data, we asked whether among differentially spliced exons the predicted targets were enriched for eCLIP peaks compared to predicted nontargets. Of the 11 considered SFs, nine showed clear eCLIP enrichment at predicted targets, with most showing 2–8-fold higher frequency of peaks compared to predicted nontargets (Fig. 3e and Extended Data Fig. 4c). These enrichments are robust across different minimum $\frac{\text{pulldown}}{\text{input}}$ cutoffs for defining eCLIP peaks (Extended Data Fig. 4d,e). Depletion of eCLIP peaks from predicted nontargets suggests that these typically reflect exons not directly bound by the SF. The positional distribution of eCLIP peaks at targets was typically consistent with the factor's activity map (Extended Data Fig. 4f,g). For example, in the QKI (HepG2) knockdown, ~70% and ~80% of differentially spliced suppression and enhancing targets had upstream and downstream QKI eCLIP peaks, compared to <5% and <15% of upregulated and downregulated predicted nontargets (Extended Data Fig. 4c,g), consistent with QKI's upstream repressing and downstream enhancing activity (Fig. 2a). Furthermore, the peaks found at predicted nontargets typically displayed a more uniform positional distribution compared to predicted targets or to all exons differentially spliced in the same direction (Extended Data Fig. 4f,g), suggestive of nonregulatory binding. To assess this, we compared the effect sizes observed at predicted nontargets with and without eCLIP peaks. For most SFs, the predicted nontargets with eCLIP peaks were no more responsive to knockdown than those without and were always less responsive to knockdown than the predicted targets (Extended Data Fig. 4h,i). This pattern indicates that most nontargets with eCLIP peaks represent nonregulatory RNA-binding events rather than true regulatory targets missed by our model. Together, these observations suggest that KATMAP's predicted targets are typically those directly bound and regulated by the knocked down SF, while the predicted nontargets more frequently reflect indirect effects of knockdown.

Predicted targets generalize across biological contexts

Knockdown and crosslinking experiments are specific to the cell types and conditions in which the experiment was performed; however, if a model successfully captures regulatory activity, its predictions should generalize to other biological contexts. To assess whether our models can generalize in this way, we considered SFs with knockdowns in distinct cell types and compared their predictions. For 12 of the 14 factors, models inferred from HepG2 and K562 cells yielded highly similar predictions of regulation (Fig. 3f), indicating that KATMAP consistently identifies the same underlying relationship between binding motifs and splicing regulation. To assess whether this generalizability extends across species, we applied KATMAP to mouse *Rbfox1/2/3* triple-knockout data[51]. The inferred activity map (Extended Data Fig. 5a) was highly consistent with those obtained from human RBFOX2 knockdowns (Fig. 2e and Extended Data Fig. 5b). Using the human RBFOX models to score exons in the mouse transcriptome predicts essentially the same regulation as the splicing impacts learned directly from mouse neuronal *Rbfox1/2/3* triple-knockout data[51] (Fig. 3g and Extended Data Fig. 5c). Together, these analyses indicate that the models KATMAP infers can be applied to different tissues and to related species, including those where in vivo binding data are lacking and where genes and exons not expressed in the original knockdown experiment are present.

KATMAP identifies *cis*-regulatory elements and the consequences of their disruption

Implicit in KATMAP's predictions are statements about *cis*-regulatory elements: for any predicted target, we can examine the collection of predicted SREs that could impact that exon's inclusion. For example, QKI is predicted to enhance the inclusion of *NF2* exon 16, which is one of two stop-codon-containing exons in the gene, the skipping of which suppresses tumor growth 3–4-fold in schwannoma[52]. Using KATMAP's outputs, the position-wise splicing impacts can be computed, which suggest that QKI enhances the exon's inclusion through a single ACUAA motif 53 nt downstream of the 5SS (Fig. 4a). Other targets are predicted to be controlled by multiple binding sites. For example, HNRNPC is predicted to suppress the inclusion of a highly conserved exon in its own 5′ UTR through multiple poly(U) motifs in both the upstream and downstream introns (Fig. 4b).

We used this interpretability to validate selected predictions using a minigene splicing assay. For five SFs (QKI, RBFOX2, DAZAP1, PCBP1 and HNRNPC), we chose predicted target exons that changed in splicing upon knockdown and evaluated the effects of mutating their predicted SREs (Supplementary Table 5). For predicted targets with both upstream and downstream SREs, we considered the effects of mutating only the upstream or only the downstream elements. We inserted the reference and mutated exons into reporters and measured exon inclusion (Fig. 4c). To confirm that the effects of mutation were mediated by the expected SF, we assayed reporters with and without knockdown of the expected factor.

Inclusion changed in the expected direction for seven of the eight mutant reporters (Fig. 4d). Mutation of the QKI-binding and RBFOX2-binding sites downstream of the *NF2* and *EPB41* exons, respectively, reduced inclusion comparable to knockdown. Disrupting the DAZAP1-binding sites upstream of the *MBNL1* exon decreased inclusion as expected, to a greater extent than the knockdown. Mutation of the PCBP1 downstream sites decreased inclusion moderately as expected but mutation of the upstream PCBP1 sites increased rather than decreased inclusion. In this case, the upstream mutations are close to the 3SS and are predicted to increase the strength of the PPT, which may override any impact of the loss of PCBP1 binding. In the HNRNPC reporters, both the upstream and downstream HNRNPC mutations behaved as expected, with either being sufficient to increase inclusion to nearly 100%. Thus, with the exception of one of the PCBP1 mutants, splicing changes always occurred in the expected direction.

To further evaluate KATMAP's *cis*-element predictions, we asked whether our models could be used to interpret two previously published tiling assays: an ASO tiling of *HRAS* exon 5 (ref. 53) and a deletion

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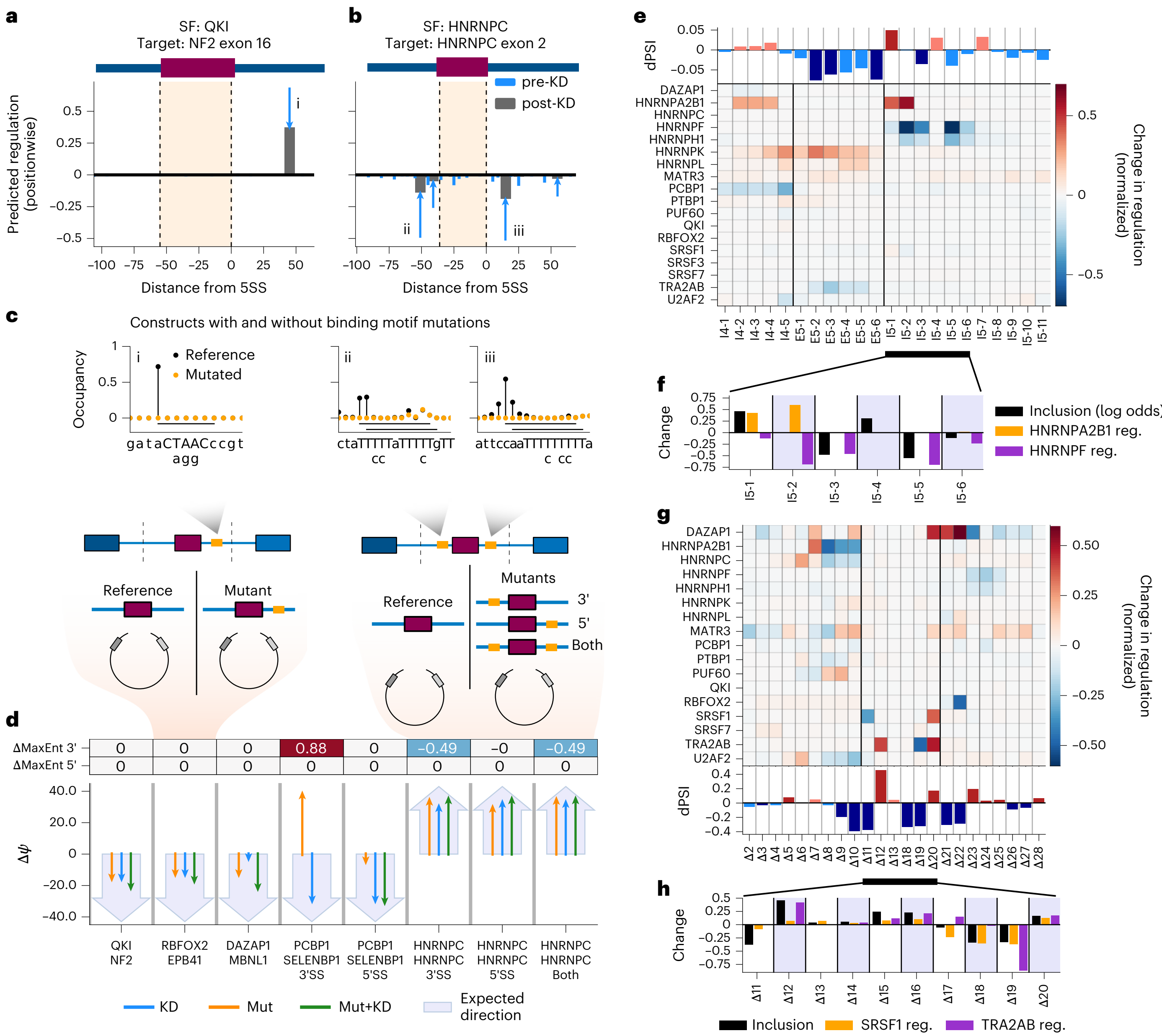

Fig. 4 | KATMAP predicts the consequences of disrupting *cis*-elements. **a**,**b**, The expected changes in regulation because of loss of QKI (**a**) and HNRNPC (**b**) binding at KATMAP's predicted targets, assuming that change in free protein was proportional to knockdown efficiency. The blue arrows indicate the change in regulation from pre-knockdown (tail of arrow) to post-knockdown (gray bar) levels. **c**, Schematics of the minigene reporter design. Top, the expected occupancy at the predicted SREs (i–iii from **a**) and the predicted effects of mutations. The reference sequence is on the *x* axis with the expected binding motif capitalized; the specific mutations are below. Bottom, schematics of the reporters constructed from these sequences. **d**, The results of the reporter assay. The thin arrows depict the observed effect of knockdown, mutation and mutation + knockdown. The large arrow depicts direction the mutations were expected to alter inclusion. **e**, The reported changes in HRAS exon 5 inclusion (top) after ASO treatments and the changes in SF regulation predicted by KATMAP (bottom). Changes in regulation are normalized to the highest activity position in each SF's activity map. Black vertical lines separate the exon from the introns. **f**, Zoomed-in view of altered inclusion and regulation by HNRNPF and HNRNPA2B1 in the downstream intron. **g**, The reported effects of deletions on inclusion of the TRA2B poison exon (bottom) and KATMAP's predicted changes in regulation (top), as in **e**. **h**, Zoomed-in view of the effect of deletions in the exon on inclusion and predicted regulation by TRA2A/B and SRSF1 based on the 200-nt exonic models.

tiling of *TRA2B*'s poison exon[54]. The *HRAS* exon tiling experiment found that ASOs targeting the downstream intron led to decreases in exon inclusion by blocking enhancing activity of HNRNPF/H proteins at G_3 and G_4 motifs[53], as well as increases in inclusion consistent with a previously reported silencing element just downstream of the 5SS, likely mediated by HNRNPA1 (ref. 55). While the HNRNPF and H1 knockdowns both yielded significant KATMAP activity maps, the HNRNPA maps failed to pass our model comparison criteria. However, the HNRNPA2B1 (K562) knockdown yielded an activity map supporting intronic repression (Extended Data Fig. 6a) and *HRAS* exon 5 was the model's highest confidence repression target; thus, it was included in these analyses.

For each SF, we predicted the changes in regulation because of ASOs blocking SREs, assuming that the ASOs disrupt recognition of their complementary nucleotides and one additional base on either side. The SFs predicted by KATMAP to be most strongly affected by the ASOs were HNRNPF and HNRNPA2B1, with HNRNPH1 showing a profile similar to HNRNPF (Fig. 4e). Two of the three ASOs predicted to disrupt

HNRNPF's enhancing activity led to decreased inclusion and four of the five ASOs predicted to disrupt HNRNPA2B1's repression led to increased inclusion (Fig. 4e,f). The exception was ASO I5-2, which was predicted by KATMAP to disrupt both HNRNPA2B1's repression and HNRNPF's enhancement and yielded no change in inclusion (Fig. 4f), perhaps because blocking both activities canceled each other out. The ability of the HNRNPA2B1 model to explain the effects of ASOs both suggests that it captures real regulatory activity and that our model comparison criterion might be excessively stringent in some cases.

Analysis of a deletion tiling experiment of the *TRA2B* poison exon again implicated expected factors. KATMAP primarily explains the effects of exonic deletions on inclusion in terms of the SR proteins TRA2A/B and SRSF1 (Fig. 4g). While deletions that increase inclusion are typically interpreted as removing silencing elements, KATMAP instead predicts that these deletions promote positive regulation by TRA2A/B by moving its binding sites closer to the SSs, where activity is higher. The *TRA2B* poison exon is particularly long (276 nt), while our activity maps only extend 60 nt from the SSs. To confirm that this was not driven by binding sites located just outside of this 60-nt window, we refitted SRSF1 and TRA2A/B models using larger windows that extend up to 200 nt into the exon (Extended Data Fig. 6b,c). These extended models infer that TRA2A/B and SRSF1 activity decays with distance from the SSs, as previously demonstrated for SR proteins[45], and similarly predict that the deletions increase regulation by moving binding sites nearer the SSs (Fig. 4h and Extended Data Fig. 6d–g). Finally, we performed a permutation analysis, exchanging $\Delta\Psi$ among deletions within the exon and within the upstream and downstream introns. The observed associations for TRA2A/B and SRSF1 were consistently stronger than the chance associations under permutation ($P = 0.003$ and $P = 0.007$, respectively).

Together, these observations support KATMAP's ability to infer *cis*-regulatory elements responsible for splicing regulation and to predict the effects of their disruption. KATMAP's predictions should, therefore, aid in predicting viable ASO targets, reducing the search space needed for screening.

Target predictions reveal cooperative regulation

Our model makes the simplifying assumption that SFs act independently of each other. However, we reasoned that we could uncover evidence of cooperative regulation by examining our model's predictions for deviations from this null assumption. To test this possibility, we focused on RBFOX and QKI, which have strikingly similar activity maps despite binding distinct motifs (Fig. 2e,f), raising the possibility of a functional relationship between them. Earlier analyses noted that the sets of exons responsive to QKI and RBFOX knockdown are overlapping[24,56,57] and observed enrichment of QKI eCLIP signal at RBFOX-downregulated exons. However, clear evidence of QKI and RBFOX binding at the same exons was not observed and it was concluded that the correlated effects of knockdown did not reflect direct coregulation by the factors[24], though another study found binding of both factors at a subset of exons[57]. These changes could arise without direct coregulation (for example, if perturbing RBFOX impacted QKI activity, altering splicing of its targets). These analyses, however, could not distinguish between the direct and indirect effects of knockdown, a limitation that might have obscured signatures of coregulation.

We used KATMAP's ability to discriminate direct regulatory targets from indirect effects to reassess the case for direct coregulation. In the scenario of strong cooperativity, knocking down either factor should primarily affect exons bound by both factors, whereas exons bound by only one should be minimally affected. As KATMAP models one SF at a time, an exon with a binding motif for the knocked down factor will be called a target, whether or not a motif for the coregulator is present. Cooperative regulation should produce a clear signature at predicted targets: knockdown-affected targets of each factor should be enriched for targeting by both coregulators compared to targets unaffected by knockdown (Fig. 5a,b). We detected this expected signature of cooperativity, with differentially spliced QKI targets being enriched for RBFOX enhancing regulation (Fig. 5c). The reciprocal relationship was also observed; predicted RBFOX2 targets affected by RBFOX2 knockdown were enriched for QKI-binding sites (Fig. 5d). Furthermore, this pattern was evident in mouse cells depleted of all three RBFOX paralogs (*Rbfox1/2/3*) (Fig. 5e), supporting that coregulation of exons by RBFOX and QKI proteins is conserved in mice (~70 million years ago).

These coregulated exons have both QKI-binding and RBFOX-binding sites (Fig. 5f), frequently multiple of each, occurring as clusters of interspersed motifs separated by a few dozen nucleotides (Fig. 5g). This proximity of binding sites suggests that QKI and RBFOX proteins could physically interact, which is supported by prior yeast two-hybrid[58] and pulldown mass spectrometry data[59]. Furthermore, reassessment of the eCLIP data focused on the predicted direct targets supports binding of both QKI and RBFOX to the same exons; QKI crosslinking was detected at both knockdown-affected and unaffected QKI targets, whereas RFBOX2 was primarily bound at the targets affected by QKI knockdown. We selected *NF2* exon 16 to directly test this, constructing reporters with the native QKI-binding and RBFOX-binding motifs mutated in all combinations and performed all combinations of QKI and RBFOX2 knockdown. We found that regulation of this exon by RBFOX2 depends on the presence of QKI's motif; while knockdown of RBFOX2 or the mutation of its motif decreased inclusion, they had no additional effect on inclusion when the QKI motif was absent (Fig. 5h). Together, these observations support direct, cooperative coregulation of exons by QKI and RBFOX proteins.

Our analyses identified 20 human exons with binding sites for both QKI and RBFOX proteins and responsiveness to either QKI or RBFOX2 knockdown (Supplementary Table 3). Of these 20, 11 are in genes involved in cytoskeleton binding and organization, with seven being actin-binding proteins (Fig. 5i). This represents strong enrichment for these categories relative to expression-matched control genes, as well as relative to the full set of RBFOX or QKI targets, consistent with a focused subprogram active only in cells where both QKI and RBFOX are expressed and distinct from their individual regulatory roles. Cytoskeletal remodeling and cell migration are integral to early neuronal development and a number of these exons occur in genes with known roles in neural and glial biology. For example, *NDEL1* is critical for cell migration during neural differentiation[60] and skipping of *NF2* exon 16 strongly promotes growth of glial tumors[52]. Both RBFOX and QKI proteins are expressed in neuronal progenitor cells, which differentiate into both neurons (where RBFOX predominates) and glia (where QKI predominates), suggesting a potential venue for coregulation[49,61].

KATMAP's models uncover the SFs responsible for splicing changes in RNA-seq data

Most knockdown experiments cause abundant splicing changes in both directions, even for established splicing activators (for example, SRSF1) and repressors (for example HNRNPL), where direct targets should be downregulated and upregulated, respectively, limiting their ability to predict causal factors. Indeed, in most ENCODE knockdowns, fewer than 30% of the significant splicing changes were predicted by KATMAP to be direct targets (Fig. 6a). Given that our previous analyses (Fig. 3c–e) showed that predicted nontargets are depleted for eCLIP peaks and typically display smaller inclusion changes, this suggests that many of the remaining splicing changes do in fact reflect indirect effects, potentially resulting from the multiday period of antibiotic selection common to knockdown experiments and the extensive crossregulation among SFs. Because KATMAP distills knockdown and RNA-seq data into transcriptomewide predictions about direct SF regulation, it can be applied to interpret splicing changes in other RNA-seq datasets following perturbations. This includes the indirect effects of knockdown, as well as the splicing changes observed following stress, drug treatments, cell differentiation and in clinical samples associated with disease.

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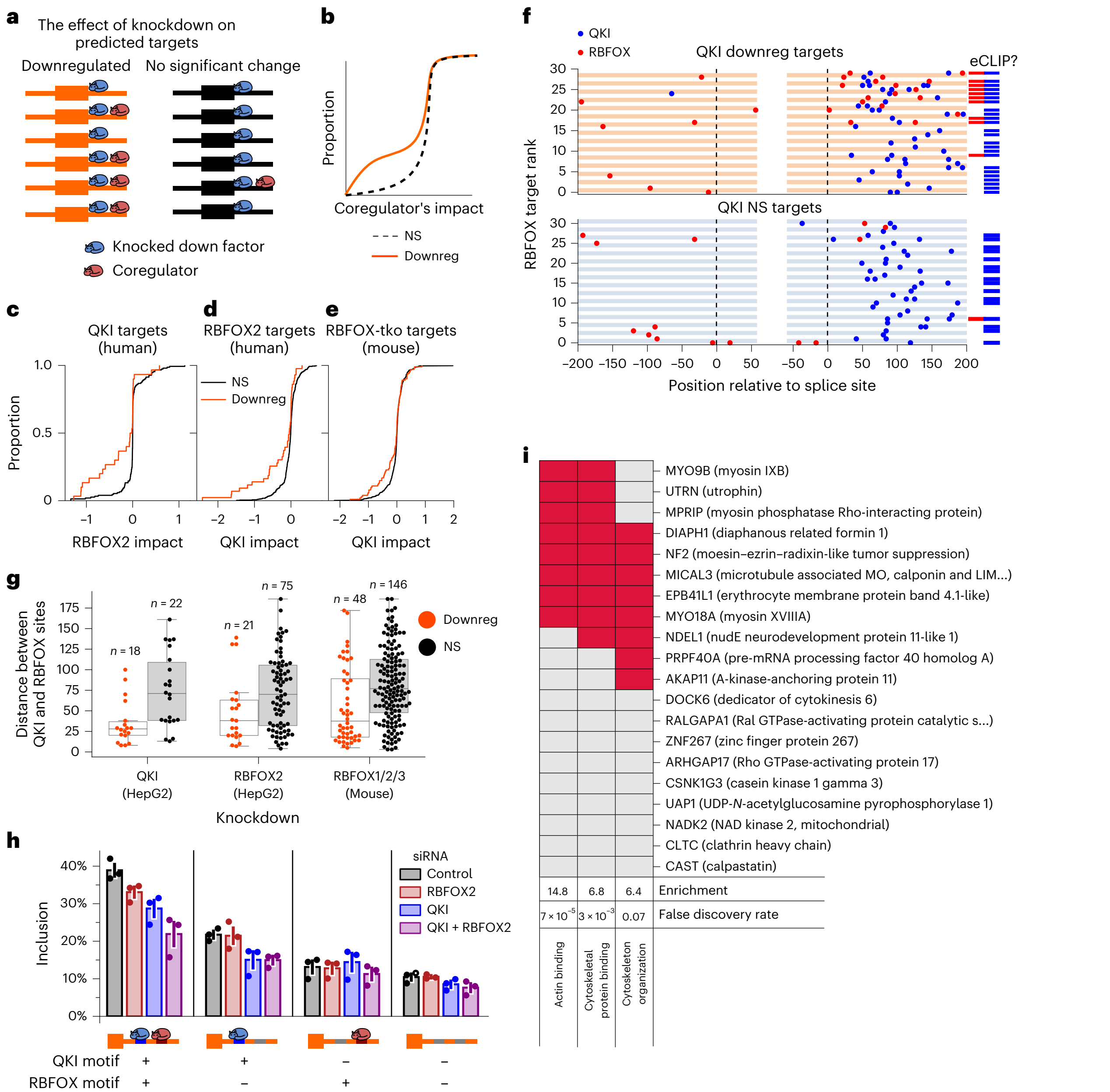

Fig. 5 | QKI and RBFOX cooperatively regulate a splicing subprogram. **a**, In the case of cooperativity, predicted targets that affected knockdown should be enriched for the coregulator compared to those unaffected by knockdown. **b**, The expected distributions of the coregulator's splicing impact computed at KD-affected (orange) and KD-unaffected targets (black). **c**–**e**, Distributions of RBFOX2 (**c**) and QKI (**d**,**e**) splicing impacts at the targets of the other protein (**c**, QKI; **d**,**e**, RBFOX), comparing the KD-affected (orange) to KD-unaffected targets (black). **f**, Locations of predicted binding sites (|relative change in binding| > 20%) for both RBFOX (red) and QKI (blue) in KD-affected QKI enhancing targets affected (**top**) and an equal number of KD-unaffected targets (**bottom**). The exons are sorted by RBFOX splicing impact and the bars to the right denote the presence of eCLIP peaks in the downstream intron. **g**, The distributions of distances between the nearest QKI and RBFOX sites in KD-affected (orange) and unaffected (black) targets. Boxes are centered on the median and span the lower and upper quartiles, with whiskers extending to the most extreme observation no more than 1.5× the IQR distant from the lower or upper quantile. **h**, Semiquantitative RT–PCR measurements of NF2 exon 16 inclusion (mean ± 1 standard error) in minigene for constructs with the QKI and/or RBFOX motifs mutated under all combinations of RBFOX2 and QKI knockdown. **i**, GO analysis of confident QKI–RBFOX cotargets; red denotes GO membership.

If a given SF contributed to the splicing changes in an RNA-seq dataset, the splicing model learned from analyses of its knockdown data should be predictive of which exons were affected. We express this logic with a multinomial logistic regression that predicts each exon's splicing changes on the basis of the splicing impacts computed using the models we learned from the different ENCODE knockdowns (Fig. 6b). To validate this approach, we used the splicing models inferred from K562 knockdowns to explain splicing changes in the HepG2 knockdowns and vice versa. This approach correctly identified the depleted SF as most predictive in 21 of 23 knockdowns (91%) (Fig. 6c), demonstrating that our regulatory models can generalize beyond their training data to infer the SFs underlying transcriptomic changes across cell types.

We asked how this approach compares to using the exon inclusion changes directly observed in other knockdown data as predictors to infer perturbations. Both KATMAP and the inclusion-based approach effectively identified the knocked down factor (Fig. 6c and Extended Data Fig. 7a). However, the two approaches disagreed as to which SFs were secondarily perturbed (Extended Data Fig. 7b).

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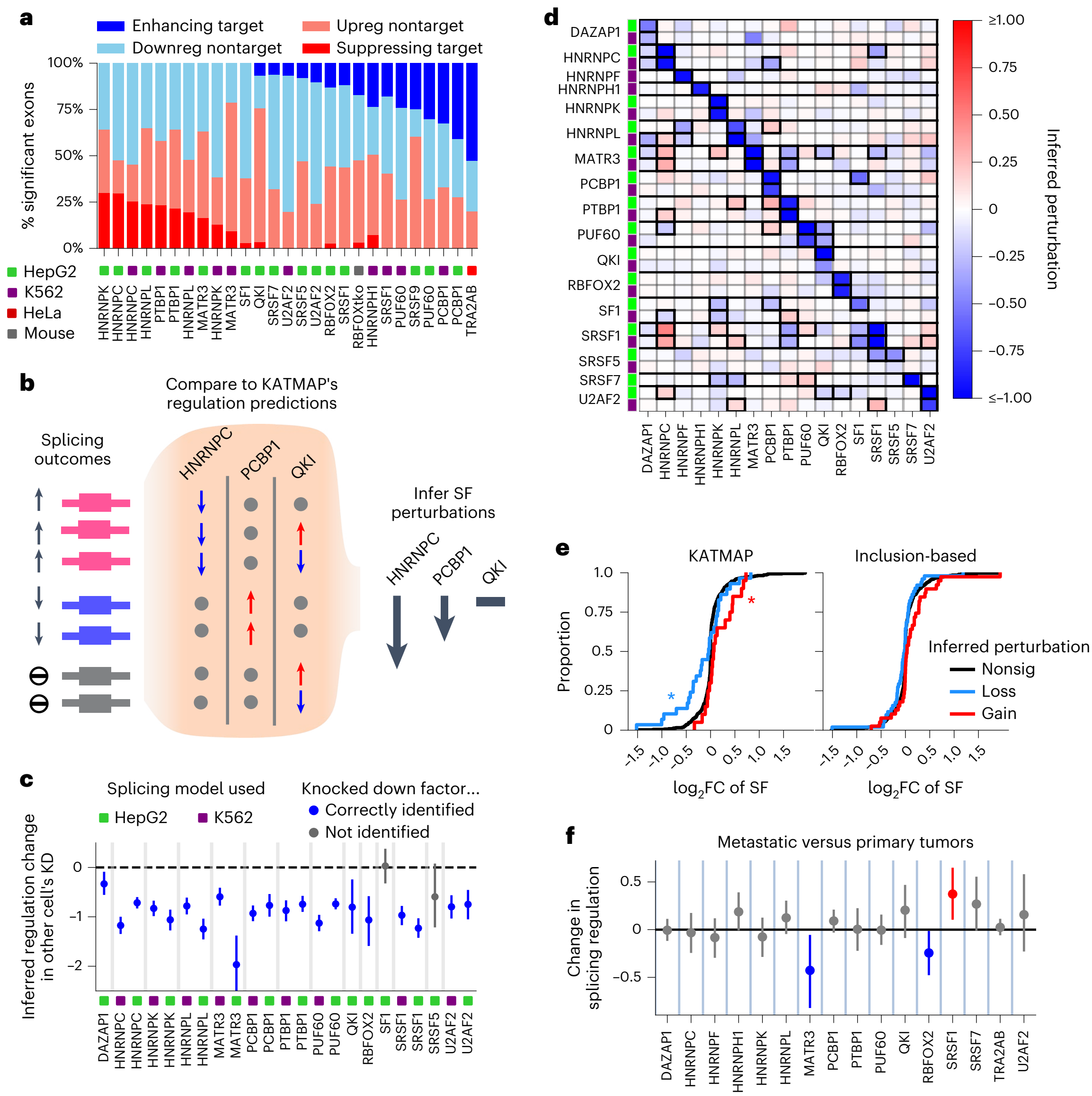

Fig. 6 | KATMAP's regulatory models uncover SF perturbations. a, The fraction of targets and nontargets among differentially spliced exons (rMATS FDR < 0.05). Ambiguous target predictions (Pr(target) and Pr(nontarget) < 0.8) are not included. **b**, Schematic showing how we infer underlying SF perturbations by comparing splicing changes to KATMAP's regulatory models. In this example, exons predicted to be suppressed by HNRNPC are upregulated and exons enhanced by PCBP1 are downregulated, suggesting that decreases in HNRNPC and PCBP1 regulation caused the splicing changes. There is no association between the splicing changes and QKI's predicted targets, suggesting no perturbations to QKI's regulation. **c**, Predicted changes in regulation for the knocked down factors ($\gamma^{(f)}$, as described in the Methods), with the 94% CI (n = 800 posterior samples). All activity models learned from the other cell type were included in the regression and knockdowns where the expected factor gave the strongest signal are indicated in blue. **d**, Predicted regulation changes for knockdowns of all RBPs with a significant activity map, using ashr to enforce sparsity. Changes significant at 10% FDR are denoted with thick boxes. **e**, The $\log_2$ fold changes in expression (DEseq, ashr shrinkage estimates) of SFs inferred to have been secondarily perturbed at a 10% FDR. *Significant according to a two-sided KS test at $P < 0.05$ ($P = 0.007$ for losses and 0.01 for gains). **f**, Predicted differences in SF regulation between metastatic and primary PDA samples, with 94% CI (n = 800 posterior samples).

To assess these disagreements, we selected significant secondary perturbations (10% false discovery rate (FDR), Fig. 6d) and asked whether these were associated with expression changes in the expected directions. The gains and losses of SF regulation inferred by KATMAP were supported by corresponding increases and decreases in expression (Fig. 6e, Kolmogorov–Smirnov (KS) statistics = 0.35 and 0.31; $P = 0.01$ and 0.007), while the gains inferred by the inclusion-based approach were more weakly supported by expression increases and not supported for inferred losses (KS statistics = 0.21 and 0.19; $P = 0.07$ and 0.06). These observations indicate that KATMAP more accurately identifies secondarily perturbed factors, likely because of its ability to focus primarily on splicing changes at the respective SF's direct targets.

MARA (motif activity response analysis)-like approaches have previously been applied to this problem; thus, we compared KATMAP to the MARA-like tool, MAPP (motif activity on pre-mRNA processing)[13]. MAPP considers a large library of RNA-binding protein (RBP)-associated PWMs or k-mers and fits linear models to identify which motifs are associated with splicing changes, simultaneously estimating impact maps that describe position-specific activity and computing absolute z scores indicating which SFs might be responsible for the splicing changes. To compare the two approaches, we considered SFs for which there were (1) knockdowns in both ENCODE cell lines; (2) PWMs in the MAPP library; and (3) a significant KATMAP model in at least one knockdown.

MAPP assigned the top *z* score to a PWM associated with the knocked down factor (or a paralog) in nine of the 24 knockdowns, while KATMAP did so in 22 of 24 (Extended Data Fig. 7c). Furthermore, because MARA-like approaches must learn the activities associated with each motif every time they are applied, MAPP struggled to distinguish factors with similar binding motifs. For example, in the HNRNPK knockdowns, MAPP identified perturbations of both PCBP1 and HNRNPK because the two SFs bind similar C-rich motifs (Extended Data Fig. 7d) and faced the same problem in the PCBP1/2 knockdowns. By contrast, KATMAP couples motifs with activity profiles previously learned from knockdown data, allowing it to distinguish between HNRNPK and PCBP1 perturbations (Extended Data Fig. 7d) because their distinct patterns of activity are accounted for (Fig. 2a). This highlights the added power gained from incorporating prior knowledge of an SF's activity when it is available.

These analyses provide a proof of principle that SF perturbations can be inferred using KATMAP's models in the context of well-controlled experiments. To assess the potential of this approach in more complex and disease-relevant contexts, we applied KATMAP to a published comparison of splicing changes between primary and metastatic pancreatic ductal adenocarcinoma (PDA) samples, which uncovered a role for RBFOX proteins in repressing metastasis[62]. Unlike knockdown data, cancer samples typically represent a mixture of distinct etiologies, which may obscure the signal of any specific causal factor. Indeed, the original analysis did not recover unambiguous RBFOX motifs, but rather amalgams of G-rich and C-rich motifs, some of which had similarity to RBFOX secondary motifs. Applying KATMAP, we directly inferred loss of RBFOX regulation in the metastatic samples through a single analysis (Fig. 6f), without the need to first identify motifs and then match them to putative SFs. We further infer a gain of SRSF1 regulation, a factor known to promote tumorigenesis[3], which was missed in the original motif analysis. The inferred perturbations to RBFOX and SRSF1 are consistent with reported protein level changes in these samples, with RBFOX protein levels decreased in the metastatic lineages and SRSF1 protein levels elevated[62]. This analysis highlights KATMAP as a powerful tool for inferring the causal factors for splicing changes in disease contexts.

Discussion

Since the development of RNA-seq, perturbation experiments have been the predominant tool for probing the rules by which regulatory factors control gene expression and processing. However, these studies typically report visual summaries of regulation, which are not directly useful for quantitatively interpreting regulation in additional biological settings. Furthermore, most analyses do not distinguish between the direct and indirect effects of perturbation, instead assuming that the signal from direct targets will rise above the background of indirect effects. Our approach, KATMAP, extracts actionable insights about splicing regulation from SF perturbation data. The activity maps that KATMAP generates are interpretable visualizations of an SF's regulatory activity and also models that can be applied to make predictions in additional contexts. These predictions distinguish the direct from indirect effects of SF perturbation, allowing indirect effects to be further interpreted in terms of secondary factors and preventing them from contaminating conclusions drawn about the perturbed factor's biological role. Our approach is supported by the reproducibility of inferred activities and predictions when applied across cellular contexts and related species. The model design makes KATMAP a general framework for rigorously applying insights from perturbation data to various types of analyses.

Despite these advantages, our approach has several limitations. Firstly, we make the simplifying assumption that each SF can be modeled individually and ignore potential cooperative regulation between SFs. Nonetheless, the regulatory models learned by KATMAP can be used as a null hypotheses to uncover evidence of cooperativity, as seen for RBFOX and QKI proteins. Secondly, KATMAP's interpretability depends on knowledge of the SF's sequence specificity. While motifs discovered de novo can be used with KATMAP, there are epistemic limitations to what can be concluded when an SF's binding motif is not known a priori. As the sequence specificity of more SFs becomes available, the set of factors for which interpretable models can be learned will expand. Lastly, while based on a general model of regulation, KATMAP is currently limited to describing splicing regulation. Recent work indicates that transcription factors also have position-specific activities[63], suggesting that modified versions of KATMAP might find application in modeling transcription.

In distilling knockdown experiments into generalizable models, KATMAP provides the opportunity to leverage decades of experimental work to answer biological questions. The insights our model extracts from perturbation data will benefit work focused on specific regulatory factors, providing information about activity and targets and quickly generating hypotheses to further probe that factor's biology. More generally, many researchers who generate RNA-seq data will perform a differential splicing analysis, typically uncovering hundreds of splicing differences, which are challenging to interpret. Integrating KATMAP into RNA-seq workflows provides a means to focus downstream analyses on factors responsible for the phenotype of interest. Our reanalyses of pancreatic cancer data illustrate a use case for inference of SF drivers of splicing changes in a clinical dataset (Fig. 6f). Identifying the SFs and SREs underlying splicing programs in disease may speed the search for therapeutic targets and lead to a better understanding of underlying causes. Furthermore, because KATMAP's predictions are explicitly framed in terms of changes in binding at *cis*-elements, our models can facilitate the interpretation of sequence variants. This is highlighted in our minigene experiments where we successfully identified regulatory sequence elements and designed mutations that disrupt that regulation. Similar analyses have applications in ASO design by providing information on which *cis*-elements are likely to change inclusion in the desired direction. Focusing on these regions can reduce the need for laborious tiling or deletion assays during therapeutic development and inference of specific SFs may aid in the design of ASOs active in relevant tissues.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41587-025-02881-9.

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